

- 1) Input values { 4371, 1323, 6173, 4199, 4344, 9679, 1989 } and a hash function  $h(x) = x \pmod{10}$ , show the resulting:-

a) Separate chaining hash table:-

Table size = 10, separate chaining

| Key  | $h(x)$ |
|------|--------|
| 4371 | 1      |
| 1323 | 3      |
| 6173 | 3      |
| 4199 | 9      |
| 4344 | 4      |
| 9679 | 9      |
| 1989 | 9      |

So, collisions occur at 3, 9, 4

| Index | chain              |
|-------|--------------------|
| 0     | -                  |
| 1     | 4371               |
| 2     | -                  |
| 3     | 1323 → 6173        |
| 4     | 4344               |
| 5     | -                  |
| 6     | -                  |
| 7     | -                  |
| 8     | -                  |
| 9     | 4199 → 9679 → 1989 |

b) Open addressing → Linear Probing formula:-

$$h_i(x) = (h(x) + i) \pmod{10}$$

~~24 x 10~~

(6173 + 1) × 10 4 occupied

(6174 + 1) × 10 5 free

$$(9679+1) \times 10 = 0 \text{ free}$$

$$(1989+1) \times 10 = 0 \text{ occupied}$$

$$(1990+1) \times 10 = 1 \text{ occupied}$$

$$(1991+1) \times 10 = 2 \text{ free}$$

Final hash table -

| index | value |
|-------|-------|
| 0     | 9679  |
| 1     | 4371  |
| 2     | 1989  |
| 3     | 1323  |
| 4     | 6173  |
| 5     | 4344  |
| 6     | -     |
| 7     | -     |
| 8     | -     |
| 9     | 1199  |

iii) Open addressing - Quadratic Probing

formula:

$$h_i(x) = (h(x) + i^2) \bmod 10$$

6173 (Since 6173 occurs in

index 3 but already as value  
already occupied)

$$(3+1^2) \times 10 = 4$$

Formula:-

$$h_i(x) = (h(x) + i^2) \bmod 10$$

insertion

4371:-

1323:-

3  $\rightarrow$  index 3

4  $\rightarrow$  index 1

6173:-

4199:-

3 occupied

9  $\rightarrow$  index 9

$$i = 1, 3 + 1^2 = 4$$

index 4

4344:-

9679:-

4 occupied

9 occupied

$$4 + 1^2 = 5 \rightarrow \text{index 5}$$

$$9 + 1^2 = 0 \rightarrow \text{index 0}$$

1989

$i = 1$ ; 0 occupied

$i = 2$ ;  $(9 + 4) \bmod 10 = 3$  occupied

$i = 3$ ;  $(9 + 9) \bmod 10 = 8 \rightarrow \text{index 8}$

| index | value |
|-------|-------|
| 0     | 9679  |
| 1     | 4371  |
| 2     | -     |
| 3     | 1323  |
| 4     | 6173  |
| 5     | 4344  |
| 6     | -     |
| 7     | -     |
| 8     | 1989  |
| 9     | 4199  |

d) Double Hashing:-

Ans:-

$$h_1(x) = x \bmod 10$$

and

$$h_2(x) = 7 - (x \bmod 7)$$

The probing formula is:

$$h_i(x) = (h_1(x) + i \times h_2(x)) \bmod 10$$

calculate  $h_2(x)$

| keys | $h_1(x)$ | $x \bmod 7$ | $h_2(x) = 7 - (x \bmod 7)$ |
|------|----------|-------------|----------------------------|
| 4371 | 1        | 5           | 2                          |
| 323  | 3        | 2           | 5                          |
| 6173 | 3        | 6           | 1                          |
| 4199 | 9        | 3           | 4                          |
| 4344 | 4        | 4           | 3                          |
| 9679 | 9        | 5           | 2                          |
| 1989 | 9        | 1           | 6                          |

Insertion:-

4371:

$h_1 = 1 \rightarrow$  index 1

1323:-

$h_1 = 3 \rightarrow$  index 3

6173:

$h_1 = 3$  occupied.

step size  $h_2 = 1$

$3 + 1 = 4$

index 4

4199:

$h_1 = 9 \rightarrow$  index 9

7344:

$h_1 = 4$  occupied

$h_2 = 3$

$4+3=7$   
 $\rightarrow$  index 7

9679:-

$h_1 = 9$  occupied

$h_2 = 2$

$9+2=1$

1 occupied.

$1+2=3$

3 occupied.

$3+2=5 \rightarrow$  index 5

1989:-

$h_1 = 9$  occupied.

$h_2 = 6$

$9+6=15 \text{ mod } 10 = 5$

5 occupied

$5+6=11 \text{ mod } 10 = 1$

1 occupied

$1+6=7$

index 7, but 7 is

occupied by 4344

next:  $7+6=13 \text{ mod } 10 = 3$

3 occupied

next:  $3+6=9$

9 occupied.

Final Result :-

| Index | Value |
|-------|-------|
| 0     | -     |
| 1     | 4371  |
| 2     | -     |
| 3     | 1323  |
| 4     | 6173  |
| 5     | 9679  |
| 6     | -     |
| 7     | 4344  |
| 8     | -     |
| 9     | 4199  |

1989  $\rightarrow$  cannot be inserted using  
 this double-hashing setup.

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